Assessing Short-Term InSAR LOS Signal and Infrastructure Exposure Around the Salt Lake City Airport Using Geospatial Big Data Techniques



Jackson North

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Abstract

This project evaluates short-term InSAR LOS (line-of-sight) signal patterns around the Salt Lake City Airport and the surrounding industrial corridor and combines these measurements with building density and land-use data to produce an integrated exposure and vulnerability assessment. A 12-day Sentinel-1 interferogram acquired between 22 October and 3 November 2025 was processed using ASF (Alaska Satellite Facility) HyP3 (Hybrid Pluggable Processing Pipeline) to produce LOS and coherence rasters. The LOS raster was converted to centimeters, cleaned, and summarized over building footprints and land-use polygons. Additional spatial context was derived from OSM (OpenStreetMap) building footprints, OSM land-use classifications, and a gridded density layer. Three map products were generated: mean LOS per building, land-use categories with LOS context, and a composite vulnerability index.

The results show a consistent spatial gradient in the LOS signal, with stronger values in the southeast industrial corridor. Although the LOS field is dominated by atmospheric variation rather than physical deformation, the exposure analysis highlights where infrastructure coincides with the strongest signal and highest density. The project demonstrates the integration of InSAR outputs, vector data, zonal statistics, and multi-criteria analysis within a geospatial big-data workflow.

1. Introduction

Salt Lake City International Airport sits within a rapidly developing industrial corridor on the west side of the Salt Lake Valley. This area contains high-value infrastructure, ongoing construction, and soil conditions influenced by lakebed sediments that can increase sensitivity to surface variability. Even small changes in ground conditions can affect runways, taxiways, rail

yards, warehouses, pipelines, and utilities. Synthetic Aperture Radar (SAR) and Interferometric SAR (InSAR) provide a way to monitor surface conditions remotely. Although single interferograms often contain atmospheric noise, they still reveal spatial patterns that can be analyzed in relation to the built environment.

This project integrates short-term InSAR LOS signal from Sentinel-1 with large spatial datasets representing buildings, land-use, and infrastructure density. The goal is to identify where strong LOS values coincide with areas of human or economic exposure and to demonstrate how SAR-derived surfaces can be used within a geospatial big-data workflow. Although the LOS raster is dominated by atmospheric variation, it remains a valid SAR measurement of signal variability and is therefore suitable for exposure and grid-based analysis in this project.

2. Problem Statement

The primary question guiding this project is:

- Can short-term LOS signal patterns around the Salt Lake City Airport be combined with land-use and infrastructure data to identify areas of elevated exposure or vulnerability?

Supporting questions include:

- Where are the strongest LOS values located within the study area?
- How do these values intersect with building footprints and land-use categories?
- Can a composite index highlight locations where LOS signal, building density, and land-use characteristics overlap?

This project does not assume that the LOS field represents true ground deformation. Because the 12-day interferogram is dominated by atmospheric phase delay, LOS is treated as a measured SAR signal rather than physical displacement. The InSAR raster is therefore used as an environmental surface that can be summarized and integrated with other big-data layers to characterize exposure patterns rather than to infer causation.

3. Data and Methods

3.1 Study Area

The study area encompasses the Salt Lake City International Airport and the surrounding northwest industrial corridor. The southern boundary follows Interstate 80, while the northern portion extends into a mixture of farmland, open land, and wetland surfaces. This range of developed and undeveloped land types provides a useful contrast when examining how radar signal behavior varies across different surface characteristics and levels of human activity.

3.2 Data Sources

The analysis relies primarily on Sentinel-1 C-band SAR data processed through the ASF HyP3 platform. A single 12-day interferometric pair acquired between 22 October and 3 November 2025 provided two key raster products: a line-of-sight (LOS) displacement layer (in meters) and an accompanying coherence raster. Both datasets were supplied at approximately 14-meter pixel spacing and were reprojected to UTM Zone 12N (EPSG 32612) for integration with other spatial layers.

Building footprints and land-use polygons were obtained from OpenStreetMap via Geofabrik. Building outlines were clipped to the AOI to allow footprint-level LOS summaries,

and land-use classes such as industrial, commercial, landfill, military, farmland, and construction were used later as categorical inputs in the composite index. A 500 meter UTM grid was created to standardize spatial summarization of LOS intensity, infrastructure clustering, and dominant land-use type across the study area. OpenStreetMap XYZ tiles served only as basemap context for visualization and were not used in any quantitative analysis.

3.3 Pre-Processing

Initial preprocessing focused on stabilizing the interferometric signal. Pixels with coherence values below 0.3 were removed to reduce the influence of unreliable phase information, particularly in water, bare soil, or densely vegetated areas. The LOS raster was then converted from meters to centimeters using a simple linear transformation *(LOS_cm = LOS_meters × 100).* After masking and conversion, LOS values across the study site ranged from approximately 1.1 to 2.9 cm, reflecting atmospheric phase delay rather than physical deformation. Raster cleaning steps removed NaN values, reduced edge artifacts, and ensured alignment with all vector datasets in the project CRS (EPSG 32612).

3.4 Spatial Summaries

Multiple tiers of spatial summarization were conducted. At the building scale, zonal statistics were calculated for each footprint, including mean LOS, minimum and maximum LOS, and the standard deviation within each polygon. These summaries captured how the radar signal interacted with individual structures.

Land-use polygons were also evaluated using zonal statistics to compare LOS behavior across different functional areas. At a broader scale, the 500-meter grid provided a uniform

spatial framework for aggregating average LOS values, estimating the density of building footprints within each cell, and identifying which land-use category dominated each grid area. This grid served as the foundational spatial unit for the composite index.

3.5 Composite Vulnerability Index

To integrate the radar-based signal characteristics with measures of human development, a composite vulnerability index was created at the grid-cell level. Three components were normalized to a 0-1 range: mean LOS intensity, building density, and categorical land-use weight. LOS values contributed more strongly where the SAR signal was locally elevated. These variations reflect atmospheric signal behavior within the interferogram rather than ground motion, so the index captures exposure to strong radar signal rather than deformation. Building density captured areas of concentrated infrastructure, while land-use categories were assigned weights according to their functional sensitivity: industrial (1.0), commercial (0.8), landfill or construction (0.6), military (0.5), and farmland or open land (0.2).

These three normalized variables were averaged to produce a final index value for each grid cell:

$$V = (LOS_norm + Density_norm + LandUse_norm) / 3$$

Index values ranged from near zero in undeveloped northern areas to approximately 0.88 in the southeastern industrial corridor. For visualization and interpretation, results were grouped into five vulnerability classes from very low to very high.

3.6 Data Architecture and Workflow Diagram

To clarify how the Sentinel-1 raster products, OSM vector datasets, and derived grid layers interact within the analysis workflow, a data-model diagram (Figure 1) summarizes the full processing pipeline. The diagram shows the transformation from raw inputs (SAR rasters, buildings, land-use, grid) into intermediate zonal-statistics tables and finally into the composite vulnerability index. This structure reflects a typical geospatial big-data architecture where heterogeneous sources are normalized, joined, and aggregated into multi-criteria outputs. The LOS raster is treated as an implicit input to zonal statistics rather than a stored table because rasters are sampled directly when generating GridStats.

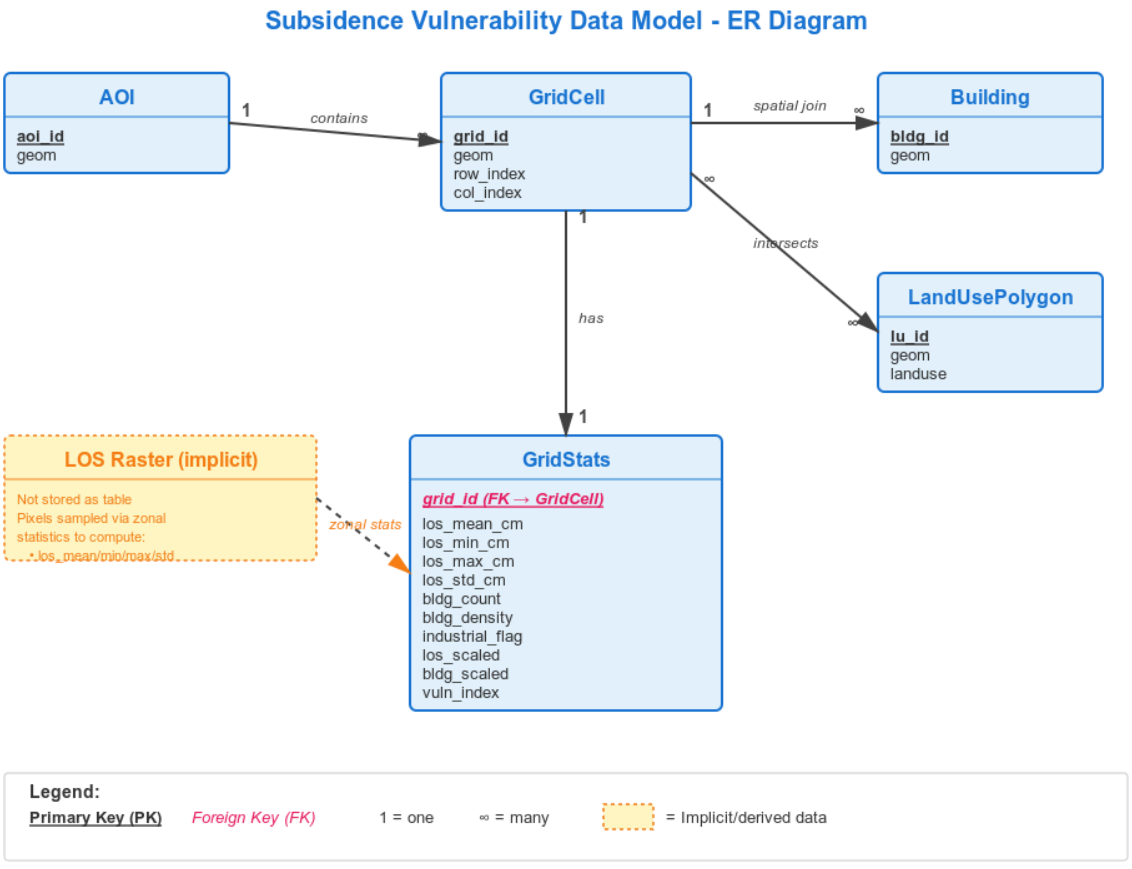
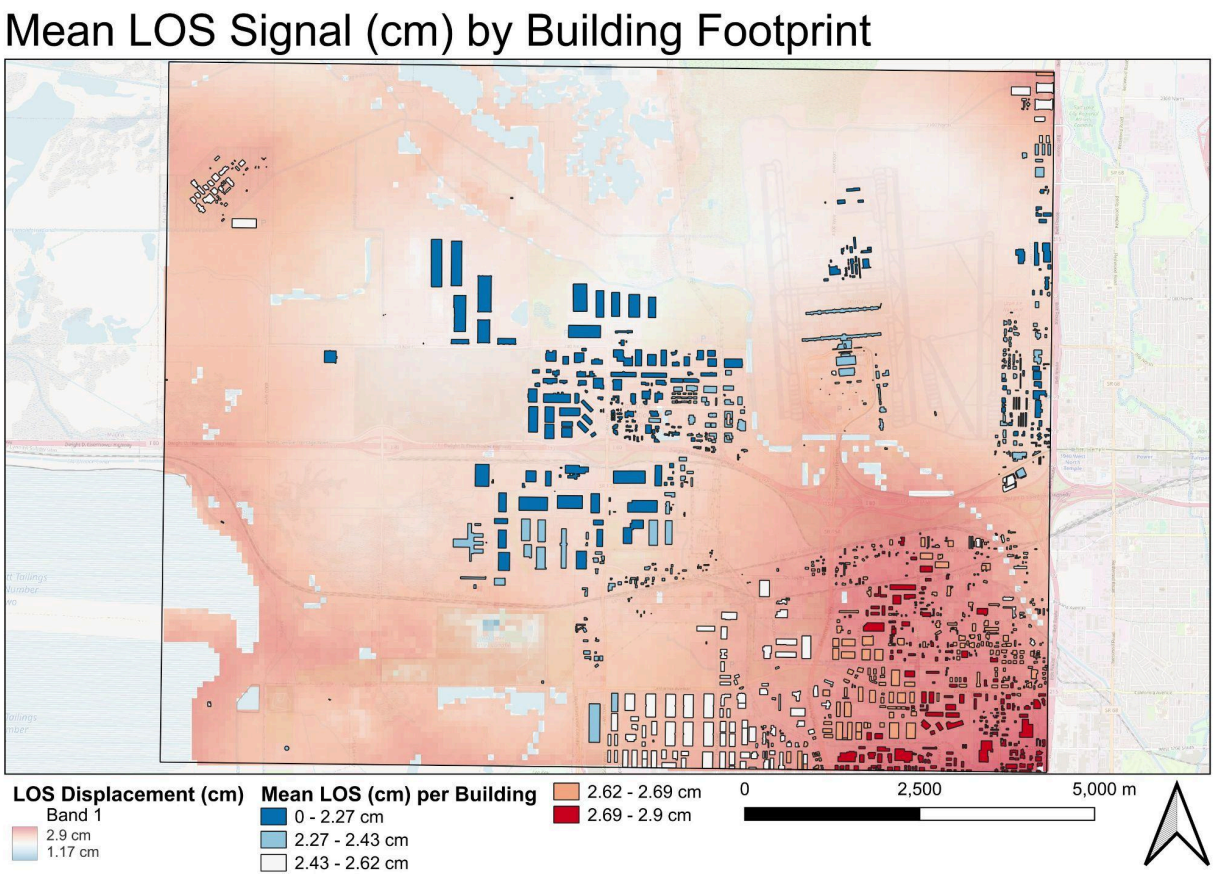


Figure 1. Entity-relationship (ER) diagram showing how Sentinel-1 LOS rasters, OSM building and land-use datasets, and the 500-m analysis grid interact to produce zonal statistics, scaled variables, and the final composite vulnerability index.

4. Results

4.1 - Figure 2 - Mean LOS Signal (cm) by Building Footprint

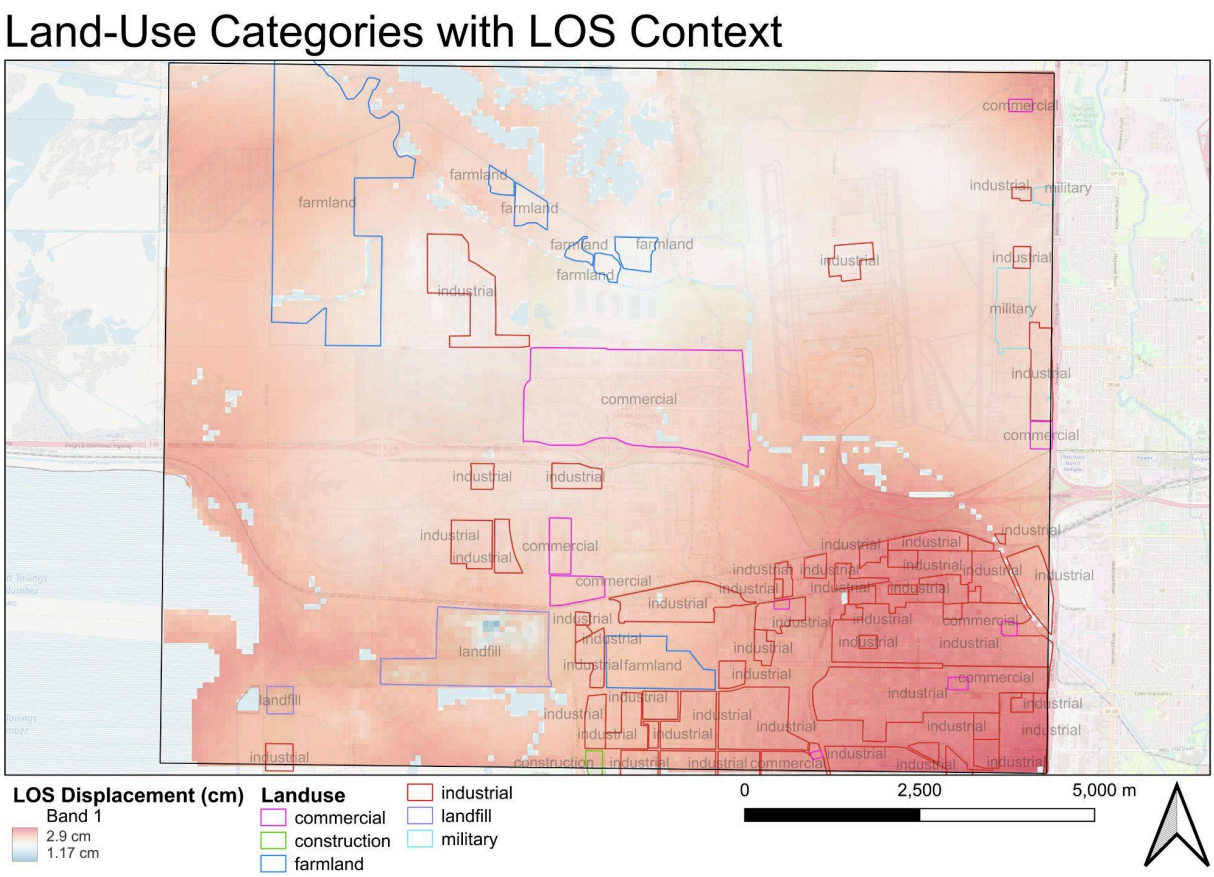


LOS summaries at the building scale reveal a clear spatial divide across the AOI. Structures in the western and northwestern areas consistently fall within the lower LOS range (1.1–2.3 cm), indicating weaker radar signal expression over farmland, open land, and lower-density commercial parcels. Moving southeast toward the industrial corridor, buildings regularly register LOS values above 2.6 cm, placing them in the highest signal classes. These

patterns remain consistent across neighboring footprints, which suggests that the LOS field is spatially coherent even though it is dominated by atmospheric phase delay.

While these values do not indicate physical deformation, the building-level summaries still identify which structures sit within zones of stronger or weaker signal. This distinction matters operationally because it shows how local surface type, infrastructure density, and radar interaction produce non-uniform signal conditions across the airport corridor. The footprint-based analysis enables exposure-type interpretation at a scale that cannot be inferred from the raster alone.

4.2 - Figure 3 - Land-Use Categories with LOS Context

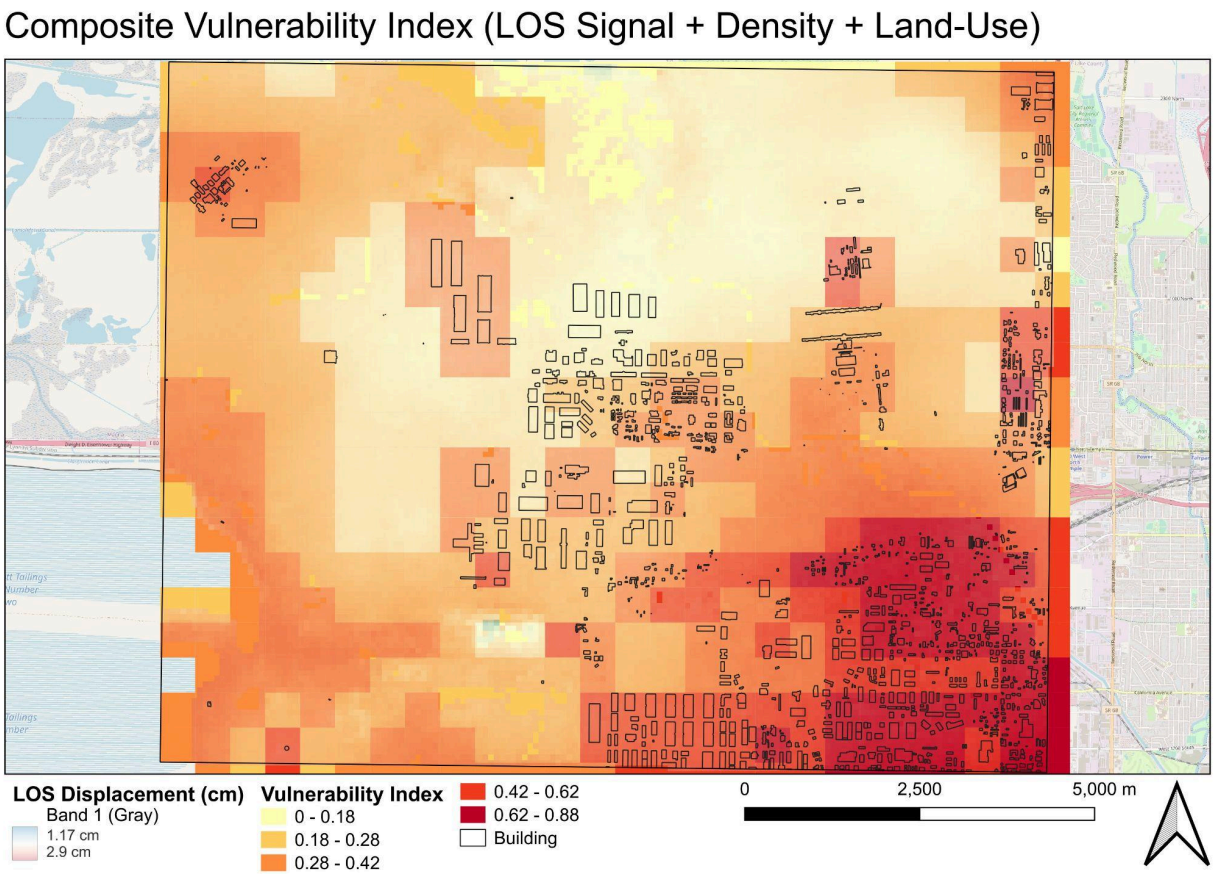


Land-use patterns align closely with the distribution of LOS intensity. Industrial polygons, which dominate the southern and southeastern AOI, overlap directly with the highest LOS signal values. These areas include clusters of warehouses, freight yards, and logistics facilities where large impervious surfaces and dense infrastructure consistently fall in elevated LOS ranges.

Commercial parcels along I-80 show a more mixed pattern, with most areas falling in mid-range LOS values. This indicates transitional zones where radar signal strength is neither minimal nor strongly elevated. By contrast, farmland and open land in the northern AOI occupy areas with the lowest LOS signal classes, matching expectations given smoother, less built surfaces and lower coherence.

Overall, this map demonstrates that LOS signal variation is not random. It reflects broad land-surface differences, and the overlay with functional land-use categories shows how environmental signal behavior intersects with human-modified landscapes.

4.3 - Figure 4 - Composite Vulnerability Index



The composite vulnerability index integrates LOS signal strength, infrastructure density, and land-use sensitivity into a single spatial metric. The highest index values occur in the southeastern industrial corridor, where all three components align: strong LOS signal, dense building footprints, and land-use categories weighted for higher exposure. These hotspots do not represent deformation risk. Instead, they identify where radar signal behavior, human infrastructure, and functional land use converge to produce zones of heightened potential impact if surface conditions were to change.

Moderate vulnerability values dominate the central corridor, where LOS values are mid-range and building density is variable. These areas represent transitional zones in both

surface type and radar interaction. The lowest values appear almost exclusively in the northern farmland and wetland areas, where both infrastructure density and LOS intensity are minimal.

The index demonstrates the value of combining heterogeneous datasets: patterns that are subtle or invisible in the raw interferogram become sharply defined when LOS, density, and land-use are analyzed together. This reinforces that the project is not evaluating deformation but is instead creating a spatial framework for understanding where multiple stressors overlap within the study area.

5. Discussion

The LOS signal field shows a clear south-to-north gradient. Based on the short interferogram, this pattern reflects atmospheric phase delay rather than physical motion. For a geospatial big-data workflow, the emphasis is not on establishing causation but on integrating spatial datasets to understand where signal strength aligns with components of the built environment.

Three patterns stand out. First, the strongest LOS values occur in the densest and most industrial parts of the study area. Second, building-level zonal statistics provide clear summaries of which structures fall within high-signal zones. Third, the composite vulnerability index highlights areas of elevated infrastructure exposure even in the absence of confirmed deformation.

This workflow meets big-data criteria because it integrates heterogeneous datasets, performs zonal summaries across many geometries, and applies grid-based aggregation and normalization to support multi-criteria evaluation.

6. Conclusions

This project shows that short-term InSAR LOS data, even when dominated by atmospheric signal, can be transformed into a meaningful analytical surface when integrated with large vector datasets in a structured geospatial big-data workflow. The LOS raster alone does not represent deformation, but once it is spatially summarized, normalized, and combined with building density and land-use characteristics, clear and interpretable patterns emerge across the airport corridor.

The results demonstrate that the strongest LOS values consistently align with industrial land use, dense development, and major transportation infrastructure. Building-level summaries reveal how radar signal behavior varies across individual structures, while the composite index identifies clusters of elevated exposure that would be invisible in the raw interferogram.

The workflow itself is the central contribution. It provides a reproducible method for turning noisy SAR surfaces into actionable spatial information through grid-based aggregation, multi-source integration, and multi-criteria evaluation. This approach can be extended to other urban and infrastructure settings where understanding atmospheric signal behavior, surface stability, or exposure patterns is more important than extracting precise deformation.

Overall, the project illustrates how SAR-derived products, vector data, and scalable spatial statistics can be combined to produce a flexible analytical framework for infrastructure assessment and spatial decision support.

References

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